

Expt 5

Power Supply

Circuits

Group no. 13

Name

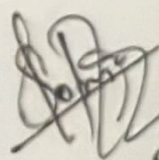
Sarvesh Muskute
Sapahazam Vishnu Singh
Saurabh Kumbhare

Roll no

220102090
220108053
220108054

Date : 26/4/23





22/05/23

- Objective
- 1) To study a full wave rectifier made using a center tapped transformer
 - 2) To study a regulated power supply made using a zener diode.

- Materials required
- 1) Equipment - Breadboard, function generator, oscilloscope
 - 2) components - Transformer (one, 230V 12-0-12V), 1N4007 (diode $V_D = 0.7V$), zener diode (one $V_Z = 6.2V$), 220 Ω , 560 Ω , 1K Ω resistors, 100 μF (one).

- Precautions & Guidelines
- 1) Switch on the main supply after making all connections.
 - 2) The capacity being of electrolytic-type is polarized & can be damaged, if connected with incorrect polarity. Similarly for diodes when connecting.
 - 3) use "line" as the source of triggering in the oscilloscope, put the oscilloscope in 'CHOP' mode.
 - 4) never ever ground the poles of CRO while measuring the voltage across transformer as it may get damaged due to short-circuit. Adjust the dc level of CRO prior to connecting the probes to the circuit.

Unregulated Power Supply - Full Wave Rectifier (FWR)

Time scale 2ms } for first graph
 Voltage per div 5V
 Phase difference = 180°

Second graph (V_o across R_L)

Time scale 2ms, 5V/div

Voltage at peak = 14V

Third graph (FWR, Full wave rectifier)

Time scale 2ms, 5V/div

Voltage at peak (V_p) = 13V Voltage at bottom peak = 12V

Ripple voltage (V_r) = 1V (13V - 12V)

4th reading (HWR, Half wave rectifier)

Time scale 2ms, 5V/div

Ripple voltage = 2V

	FWR	HWR
V_r	0.96V 0.96V	1.95V 1.95V

PART B :- Regulated Power Supply - Zener Regulator

Time Scale 1ms

R_L	$V_o(max)$	V_r	$V_{or}(max)$	V_{rR}	$V_o(avg)$	$V_{or}(avg)$
∞	13V	0.9V	5.5V	5mV	12.55V	5.5V 5.4975V
1K Ω	13.5V	0.9V	5.4V	6mV	13.05V	5.397V
220 Ω	12.8V	1.08V	3.7V	0.32V	12.26V	3.54V
R_L	I_R	I_L	I_Z	P_Z	P_R	% Reg
∞	12.6mA	0	12.6mA	69.27mW	88.86mW	0%
1K Ω	13.66mA	5.397mA	8.27mA	44.63mW	104.58mW	1.83%
220 Ω	15.57mA	16.09mA	0.52mA	1.84mW	120.81mW	5.81%

Done ✓
 24-04-2023.

(avg)
99.13

Q1 Ans : Yes, as we

Q1 Ans : NO, we didn't get the peak voltage same in the both halves.

$$\text{Top peak} = 13V$$

$$\text{Bottom peak} = 12V.$$

Because the centre tap is not proper.

$$\text{ripple frequency in FWR} = \frac{1}{3.8 \times 5 \times 10^{-3}} = 52.63 \text{ Hz}$$

$$\text{" " in HWR} = \frac{1}{5 \times 2 \times 10^{-3}} = 100 \text{ Hz}$$

Q2 Ans : Yes, as we know for voltage regulation being low the performance is better. poor regulation occurs due to low value of $R_L (= 220\Omega)$.
Explain properly.

Q3 Ans : $I_Z(\text{min}) = 5 \text{ mA}$

$$P(\text{max}) = \frac{1}{2} \text{ W}$$

$$V_{\text{max}} (\text{across zener}) = \frac{P_{\text{max}}}{I_Z(\text{min})} = \frac{0.5}{5 \times 10^{-3}} = 100V.$$

$$I_R = \frac{V_0 - V_{OR}(V_Z)}{R} = \frac{12.8 - 3.7}{220} = 41.36 \text{ mA}$$

560 Ω

We can modify the circuit by providing the higher value of R_L .

$$P_Z = 0.5 \text{ W}$$

$$V_Z I_Z = 0.5 \text{ W}$$

$$I_L(\text{max}) = I_R - I_Z(\text{min})$$

Take series with less voltage.

$$I_L (\text{max}) = 41.36 - 5 \\ = 36.36 \text{ mA} \quad \checkmark$$

Q4 Ans: P wattage ratings.

$$P = 4.2 \quad P = (41.36)^2 \times 220 \times 10^{-6} \text{ W}$$

$$P = 0.376 \text{ W} \quad ??$$

Yes, we have used correct wattage resistance
as max power dissipated in our resistance,
i.e., the wattage rating is less than rating
of resistor

Conclusion?

EE 102 Lab report

Experiment - 4

DIODE CIRCUITS

L4 Table 13

Name

Roll No

Sarvesh Murkute

220102090

Sapaharam Vishnu Singh

220108053

Saurabh Kumbhare

220108050

Ceptrum

27/04/23

Hi junior

How are



OBJECTIVE:-

- To Study half-wave rectifier circuits and
- and output capacitor filter.
- To Study the behaviour of clipping circuits.

MATERIALS REQUIRED:-

- Breadboard, Function generator, Oscilloscope
- 1k Ω Resistor, Zener diode (4V $\rightarrow$ 9V), 5k Ω , 10k Ω , 22k Ω , 47k Ω
- 22 μ F

PRECAUTIONS AND GUIDELINES:-

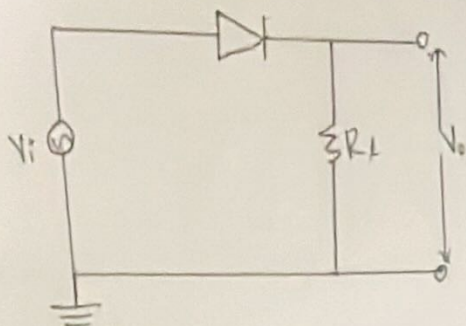
- Connect the capacitor with correct polarity. Confirm polarity of diodes before connecting
- Keep the ground terminals of oscilloscope probes and Function generator outputs connected together throughout the experiment
- Increase Sensitivity in oscilloscope to better resolution

Experiment - 4

Poe-Lab

Group - 13

8



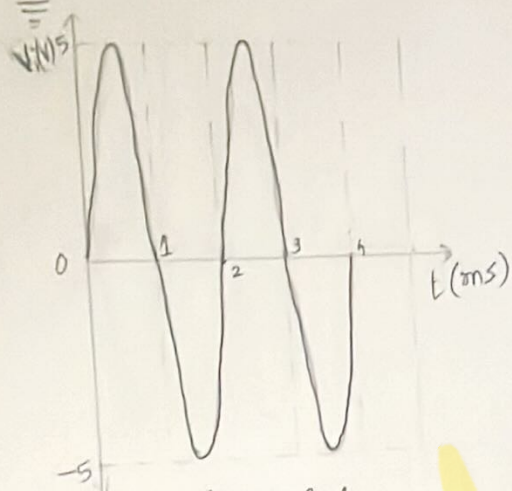
$$V_i = 10V (PP)$$

$$f = 500Hz$$

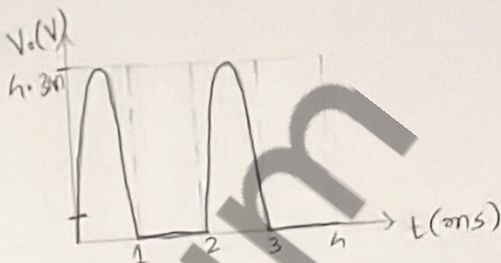
$$R_L = 2.2K\Omega$$

$$V_p = 0.7V$$

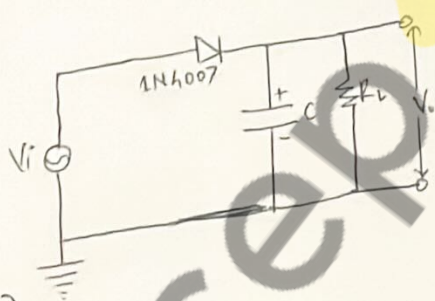
$$V_o (expected) = 8.6V (PP)$$



channel 1



channel 2



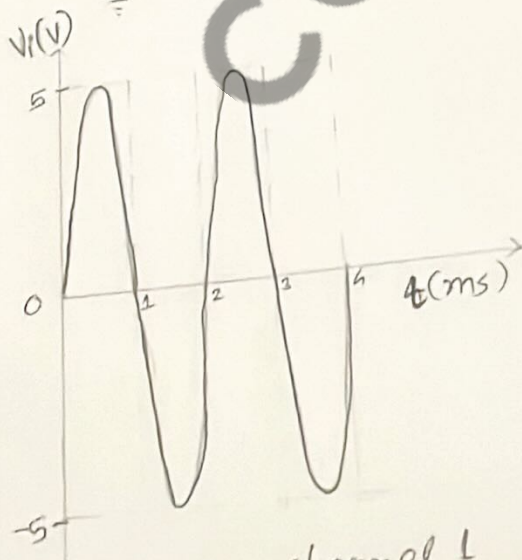
$$V_i = 10V$$

$$f = 500Hz \Rightarrow T = 2ms$$

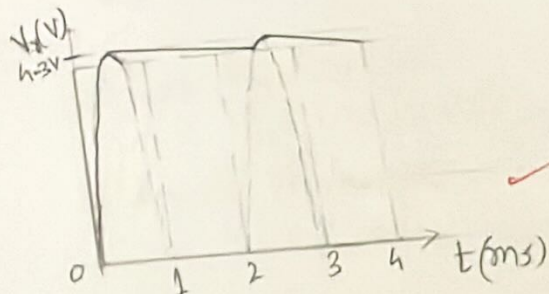
$$R_L = 2.2K\Omega, C = 22\mu F$$

$$V_{dc} = \left(1 - \frac{T}{2R_L C}\right) 5 = 4.896V$$

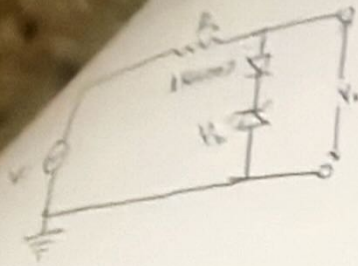
$$V_o (PP) = \frac{T}{R_L C} \times 5 = 0.208V$$



channel 1



channel 2



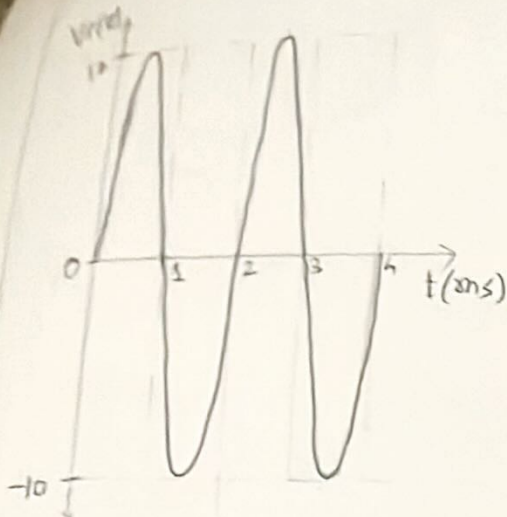
$$V_i = 20V \text{ (PP)}$$

$$f = 500 \text{ Hz}$$

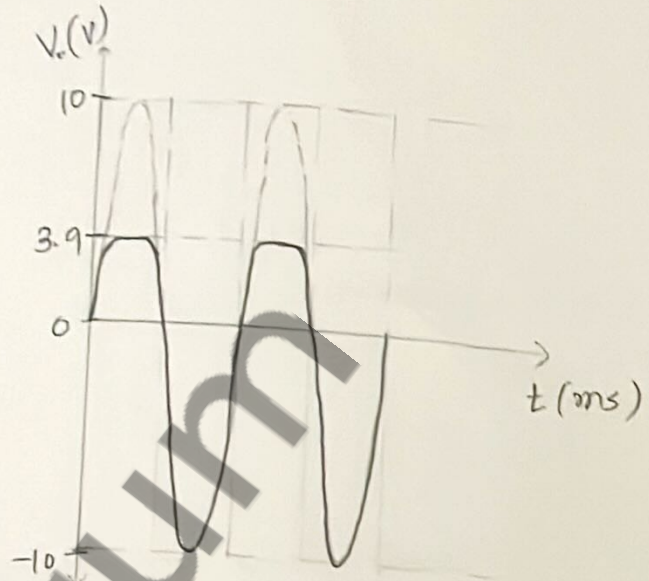
$$V_D = 3.9V$$

$$R = 2.2k\Omega$$

$$V_A = 0.7V$$



channel 1



channel 2

$$V_o - V_i = L$$

Anchal
Mawya

group-13 (LH-B)

Saurabh Kumbhare
- 220108054

Sarvesh Muskatle
- 220102090

Vishnu Singh
- 220108053

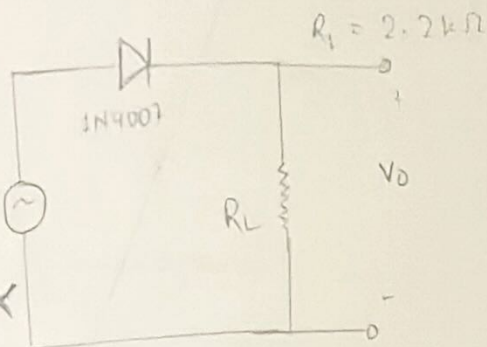
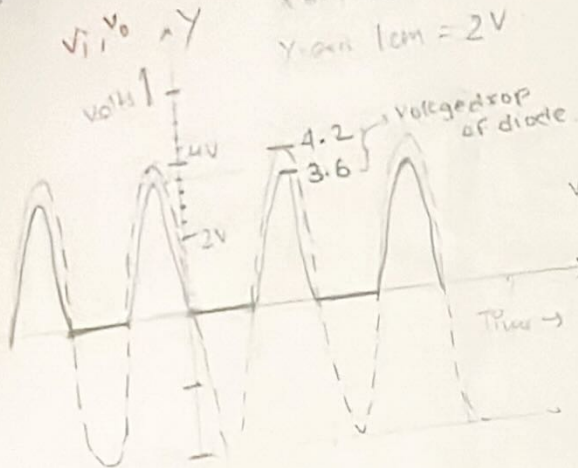
HALF WAVE RECTIFIER WITHOUT FILTER

$$V_i = 10 \sin(2\pi \times 500t) = 10 \sin(3141.6t)$$

Scale

X-axis 1cm = 1ms

Y-axis 1cm = 2V



Cutoff voltage of diode = 0.7V

Diagram

--- $\Rightarrow V_i$ (input)
— $\Rightarrow V_o$ (output)

- We can ~~exp~~ observe the output as a halfwave (V_o).
- Peak ~~to peak~~ voltage of input wave = 4.2V
- Peak voltage of output wave = 3.6V
- Difference of their peaks = 0.6V = voltage drop across diode

PART-B : HALF WAVE RECTIFIER WITH CAPACITOR FILTER

$$V_i = 10 \sin(2\pi \times 500t) = 10 \sin(3141.6t)$$

Scale:-

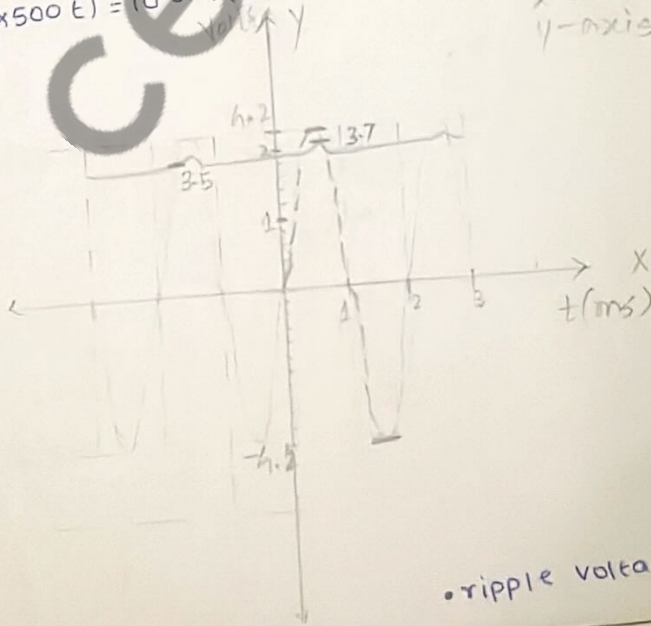
X-axis 1cm = 1ms
Y-axis 1cm = 2V

Case 1:-

$R_L = 22k\Omega$

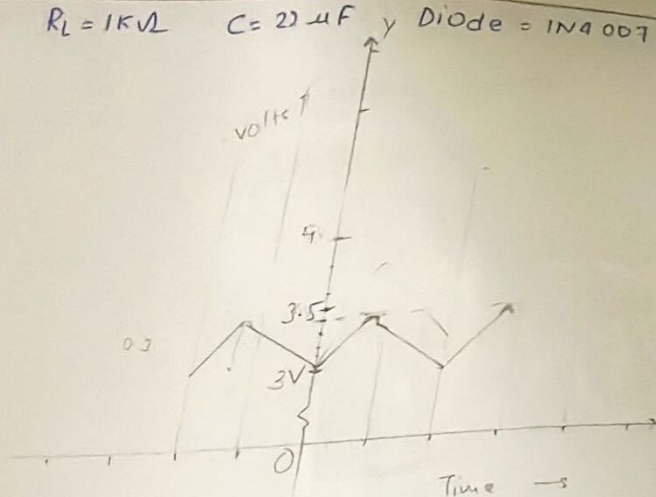
$C = 22\mu F$

Diode = 1N4007



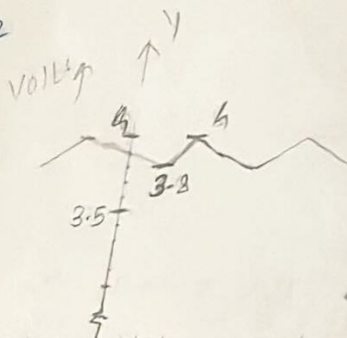
• ripple voltage = 0.2V

Case II :- $R_L = 1K\Omega$ $C = 20\mu F$ Diode = 1N4007



ripple = 0

Case III :- $R_L = 4.7K\Omega$



scale

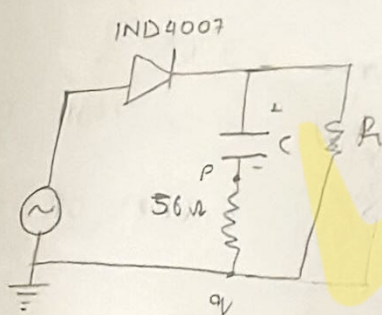
x-axis $1cm = 1ms$
y-axis $1cm = 0.5V$

ripple = 0.2V

x-axis

t (ms)

Case IV :-



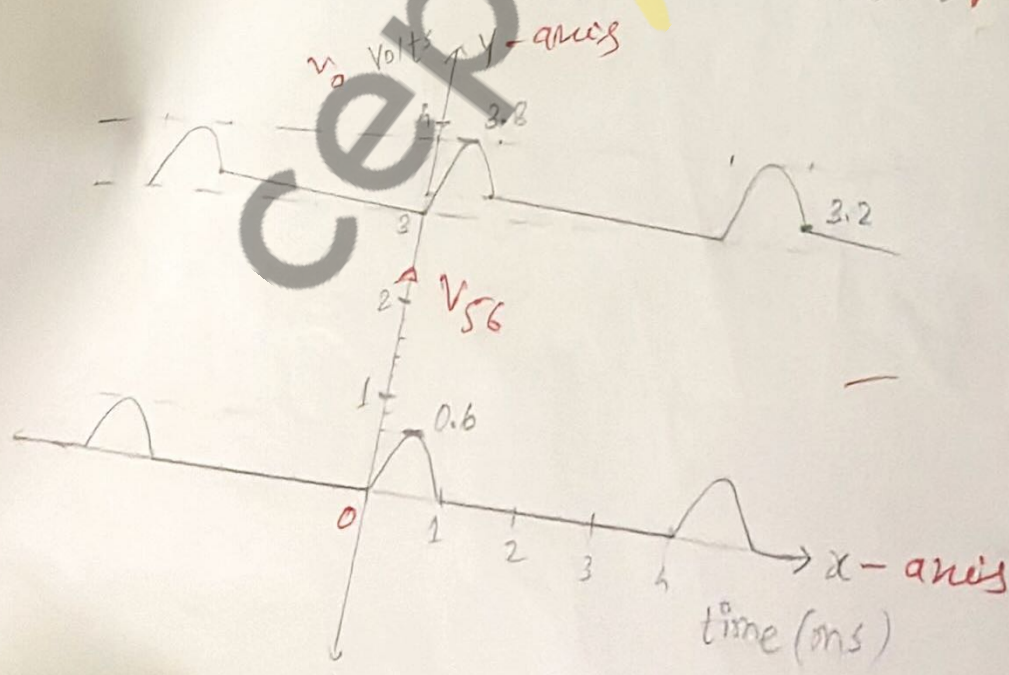
scale:

x-axis - $1cm = 1ms$

y-axis - $1cm = 1V$

Use

graph paper

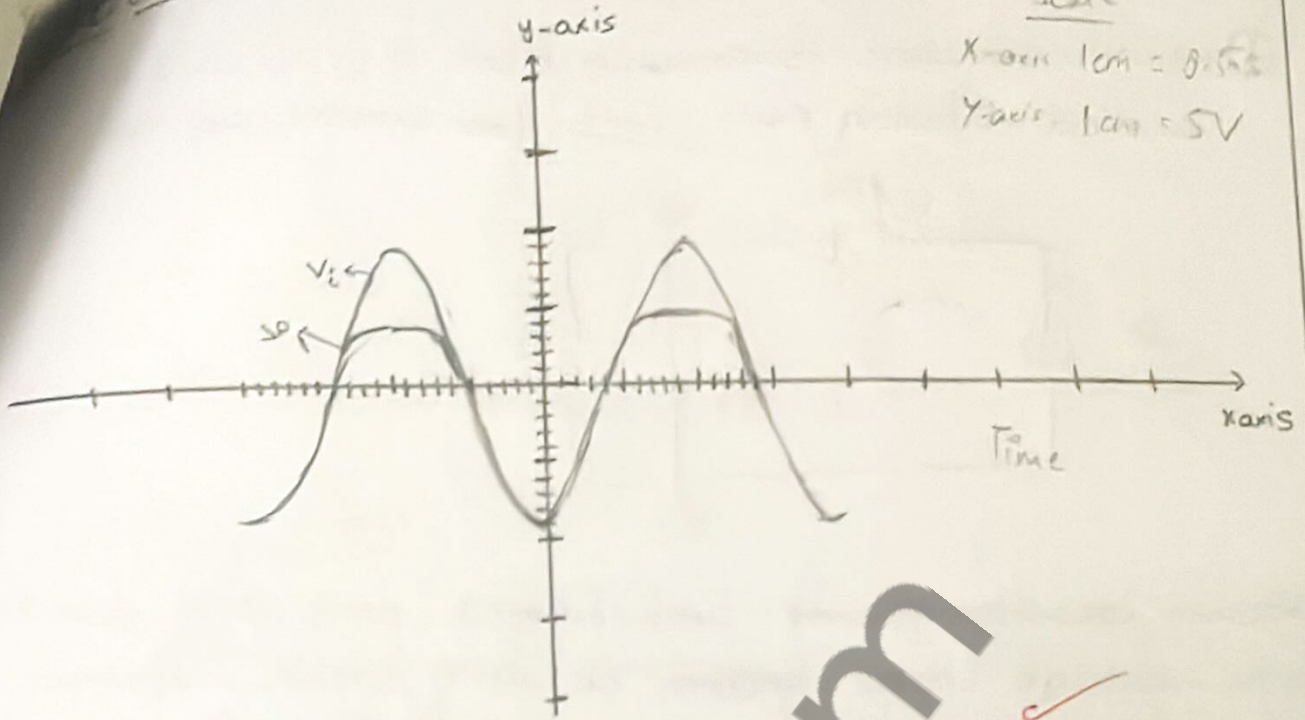


CLIPPING CIRCUIT - POSITIVE CLIPPER

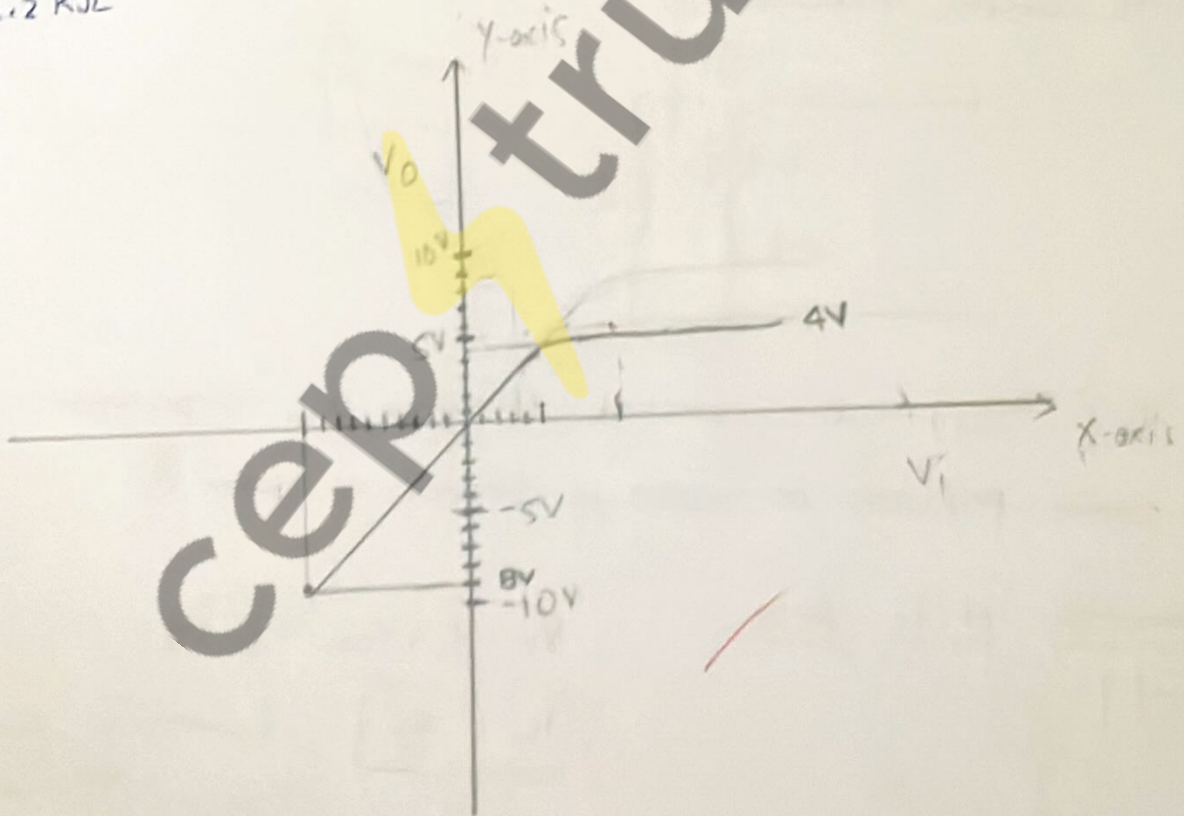
Scale

X-axis 1cm = 0.5ms

Y-axis 1cm = 5V



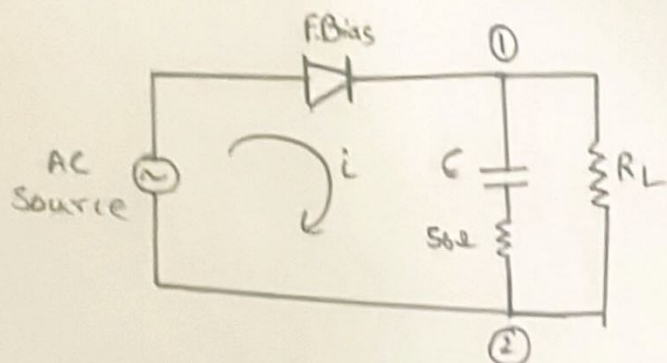
$$R_L = 2.2 \text{ K}\Omega$$



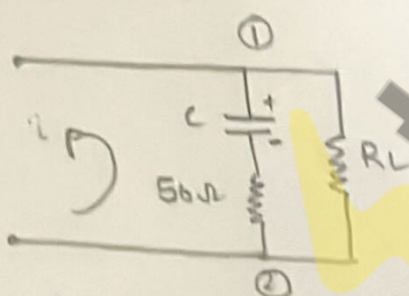
Anchal
Mawya

PART-D:- LAB REPORT

- ① (a) When capacitor is charging diode is in forward bias, current flowing from 1 → 2 (ve current direction)

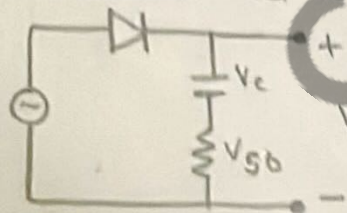


- (b) When capacitor become fully charged, diode is in reverse bias, voltage source become an open circuit, capacitor start discharging, so current is flowing from 2 → 1 (-ve current direction)



So $I_C (V_{S6}/S6) \rightarrow$ the current through the capacitor, negative for some portion of cycle is shown in part (b)

Estimation of I_C peak :-



$$V_o = V_c + V_{S6} = i_c \times Z$$

$$i_c = \frac{V_o}{Z}$$

$$i_c = \max \Rightarrow V_{o\max} \Rightarrow V_{i\max} - 0.7$$

$$V_{o\max} = 5V - 0.7V = 4.3V$$

$$Z = (X_c^2 + R^2)^{0.5}$$

$$Z = 57.84 \Omega$$

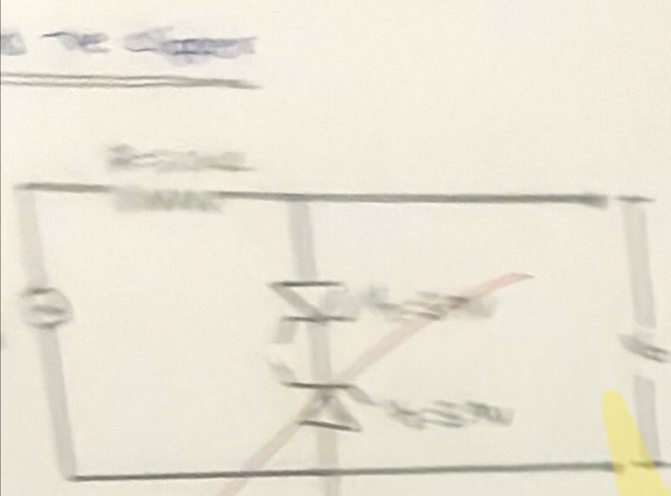
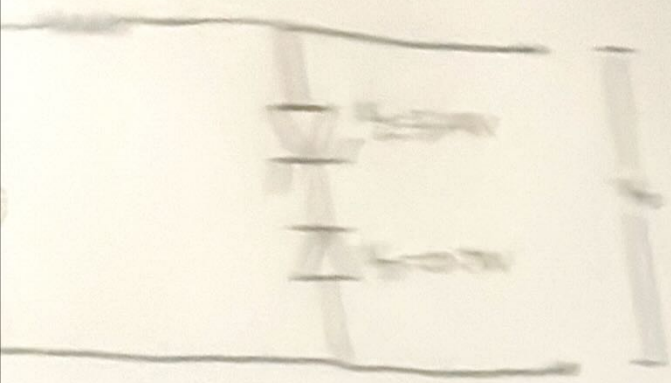
$$X_c = \frac{1}{\omega C} \Rightarrow X_c = 19.46 \Omega$$

$$C = 22 \times 10^{-6} F \quad R = 56 \Omega$$

$$\omega = 2\pi \times 500 \times 10^3 \pi$$

$$i_{c\max} = \frac{V_{o\max}}{Z} = \frac{4.3}{57.84} A = 0.0743 A$$

$$i_{c\max} = 0.0743 A$$



Theory??
Conclusion??

Ceptrum

EE-102

LAB REPORT

EXPERIMENT-III

VERIFICATION OF CIRCUIT THEOREMS

TABILE-13

L4B

NAME

1) SARVESH MURKUTE

2) SAMPAD VISHWAS

3) SAURABH KUMBHARE

ROLL NO

220107090

220108053

220108054

cep  trum

OBJECTIVE :-

To verify the Superposition Theorem, thevenin's theorem and Maximum Power transfer Theorem.

MATERIALS REQUIRED :-

- Bread board
- Multi-Output DC power supply
- Oscilloscope
- 10Ω , 560Ω , $1k\Omega$ (3), $2.2k\Omega$ (3), $3.9k\Omega$ (1), $4.7k\Omega$ (1) resistors

PRECAUTIONS AND GUIDELINES :-

- Make sure the ground terminals of the oscilloscope probe and power supplies are connected together in the circuit.
- While switching on the set-up, switch on the oscilloscope first, followed by the power supply.

PRE LAB WORK :-

- Attached in next page.

LAB READINGS :-

- Attached after pre lab

Not need to write

Sarvesh Murkute

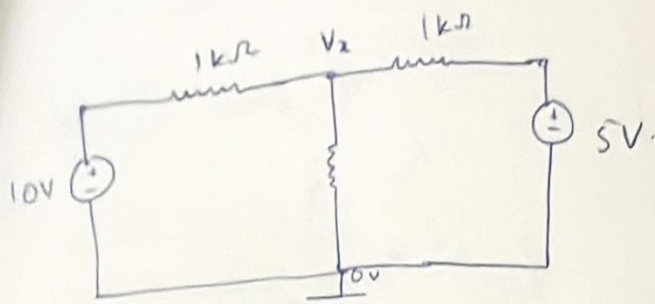
Group 13

220102030

10

EE 102

Experiment-2 (Pre-Lab work)



Applying nodal analysis

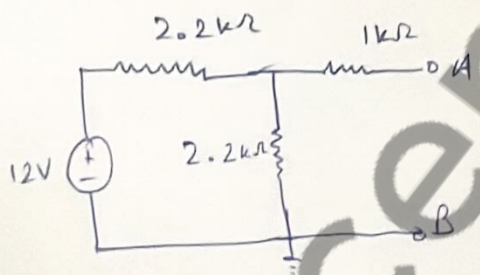
$$\frac{V_x - 10}{1} + \frac{V_x}{1} + \frac{V_x - 5}{1} = 0$$

$$3V_x = 15$$

$$V_x = 5V$$

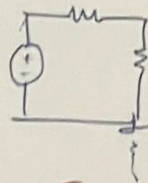
$$V_x = 5V$$

2)

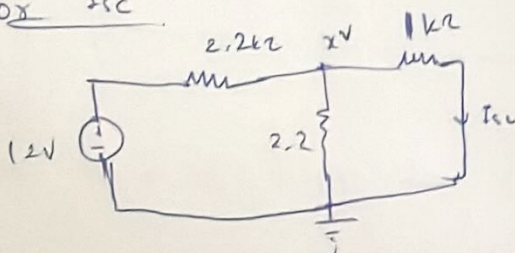


$$V_{Th} = \frac{12 \times 2.2}{2.2 + 2.2}$$

$$V_{Th} = 6V$$



for I_{sc}



$$\frac{x}{1} + \frac{x}{2.2} + \frac{x - 12}{2.2} = 0$$

(nodal analysis)

$$x + \frac{x}{1.1} = \frac{12}{2.2}$$

$$2.1x = 6$$

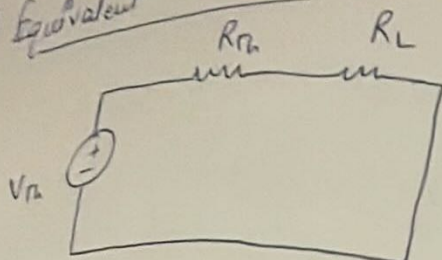
$$x = \frac{6}{2.1}$$

$$I_{sc} = \frac{6}{2.1}$$

$$R_{Th} = \frac{6}{6/2.1}$$

$$\Rightarrow R_{Th} = 2.1 \text{ k}\Omega$$

Equivalent circuit



for max power transfer

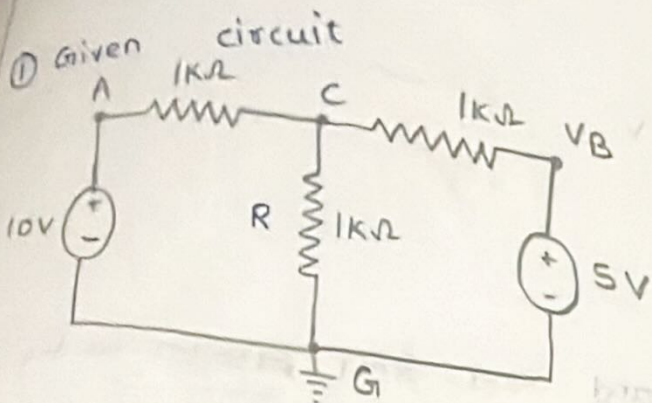
$$R_L = R_{Th}$$

$$P_{max} = \left(\frac{V}{2R_{Th}} \right)^2 \cdot R_{Th} = \frac{V^2}{4R_{Th}}$$

$$P_{max} = \frac{(6)^2}{4 \times 2.1}$$

$$\Rightarrow P_{max} = 4.285 \text{ mW}$$

Experiment 2 :- Verification of Circuit Theorems
PRE-LAB WORK



$$V_A = 10V$$

$$V_B = 5V$$

Applying KCL at C

$$\frac{V_A - V_C}{1k} + \frac{V_B - V_C}{1k} = \frac{V_C - 0}{1k}$$

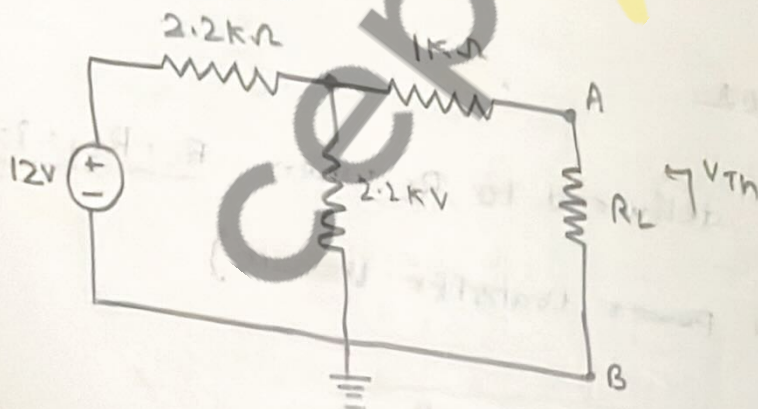
$$15 - 2V_C = V_C$$

$$15 = 3V_C$$

$$V_C = 5V$$

∴ Voltage $V_C = 5V$ and Voltage across resistance R is $V_C - V_G(V_C - 0) = V_C = 5V$

② Given circuit

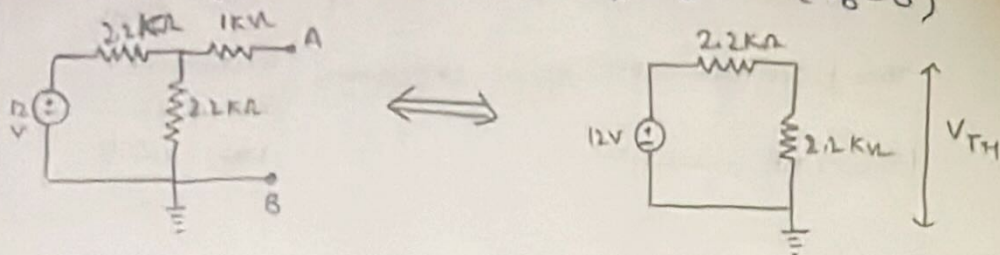


We need to find V_{TH} and R_{TH} .

V_{TH} → Thevenin's Voltage

R_{TH} → Thevenin's Resistance

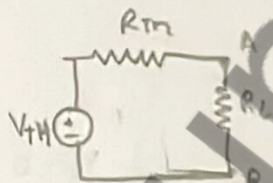
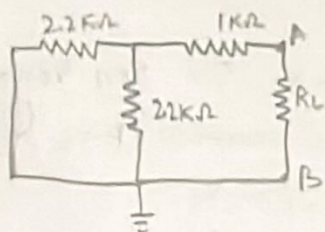
For V_{TH} ignore R_L and find $V_A - V_B$ ($V_B = 0$)



$$V_{TH} = \frac{12 (2.2)}{2.2 + 2.2} V = 6V$$

$$V_{TH} = 6V$$

For R_{TH} short circuit 12V and find R_{eff} ~~parallel~~ to R_L Series



$$R_{TH} = \frac{1}{\frac{1}{2.2} + \frac{1}{2.2}} k\Omega + 1k\Omega \quad \left(\begin{array}{l} \text{Two } 2.2k\Omega \text{ resistances are in} \\ \text{parallel and their combination} \\ \text{is in series with } 1k\Omega \end{array} \right)$$

$$R_{TH} = 1.1k\Omega + 1k\Omega$$

$$R_{TH} = 2.1k\Omega \rightarrow 2100\Omega$$

Maximum power is delivered to R_L when $R_L = R_{TH} = 2.1k\Omega$

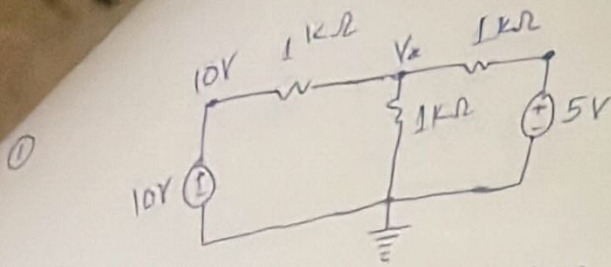
$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} \quad (\text{Maximum power transfer theorem})$$

$$= \frac{6^2}{4 \times 2.1} \times 10^{-3} = 4.285mW = P_{max}$$

∴ Maximum power that can be ~~trans~~ delivered to R_L is 4.285mW and Thevenin's voltage (V_{TH}) is 6V and thevenin's resistance (R_{TH}) is 2.1kΩ (2100Ω)

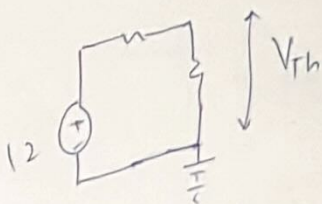
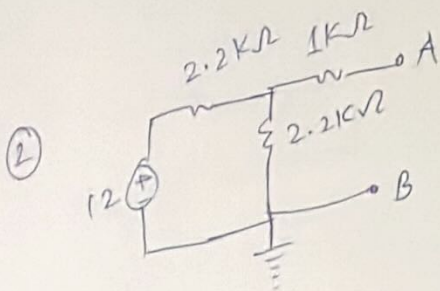
Experiment 2 (pre-lab work)

Saurabh Kumbhare
Roll No:- 220102054

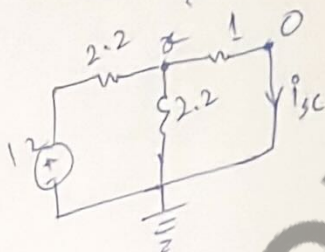


applying Nodal analysis:-

$$\frac{V_x - 10}{1} + \frac{V_x}{1} + \frac{V_x - 5}{1} = 0 \Rightarrow 3V_x = 15 \Rightarrow V_x = 5V$$



$$\Rightarrow V_{TH} = \frac{12(2.2)}{(2.2+2.2)} = 6V$$



$$\Rightarrow \frac{x}{1} + \frac{x}{2.2} + \frac{x-12}{2.2} = 0$$

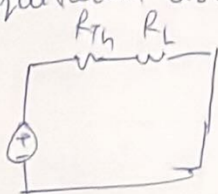
$$\Rightarrow x + \frac{x}{1.1} = \frac{12}{2.2} \Rightarrow x = \frac{6}{2.1}$$

$$\Rightarrow I_{sc} = \frac{6}{2.1}$$

$$\Rightarrow R_{TH} = \frac{6(2.1)}{6} = 2.1k\Omega$$

$$\Rightarrow R_{TH} = 2.1k\Omega \text{ \& } V_{TH} = 6V$$

Equivalent circuit:-



max. power transfer through R_L is when

$$R_L = R_{TH}$$

$$\Rightarrow \left(\frac{V}{2R_{TH}}\right)^2 \cdot R_{TH} = \frac{V^2}{4R_{TH}} = \frac{(6)^2}{4 \times 2.1} = 4.285mW$$

of Superposition theorem

	V_1 (Volt)	V_2 (Volt)	V_A (Volt)	V_B (Volt)	V_C (Volt)
Case - I	10V	5V	9.8V	4.8V	4.8V
Case - II	10V	0V	10V	0V	3.4V
Case - III	0V	5V	0V	5V	1.6V

PART - B

Thevenin's theorem

V_{OC} (Volt)	$V_{L(10\Omega)}$ (Volt)	V_{TH} (Volt)	R_{TH} (Ω)
5.9V	28mV	5.9V	2100 Ω 2143 2107 Ω

PART - C

Maximum power transfer theorem

Anchal
Marwaha
29/03/23

R_L (Ω)	V_A (Volt)	I_L (A)	V_C (Volt)	Power (Watt)
560	1.25	2.25×10^{-3}	22.5×10^{-3}	2.83×10^{-3}
1000	1.95	1.9×10^{-3}	19×10^{-3}	3.61×10^{-3}
2200	3	1.35×10^{-3}	13.5×10^{-3}	4×10^{-3}
3900	4	1×10^{-3}	10×10^{-3}	3.9×10^{-3}
4700	4.1	0.85×10^{-3}	8.5×10^{-3}	3.4×10^{-3}

PART A :- SUPERPOSITION THEOREM

Write about
three cases

$$V_{\text{case I}} = 4.8\text{V}$$

$$V_{\text{case II}} + V_{\text{case III}} = 3.4 + 1.6 = 5.00\text{V}$$

$$\% \text{ error} = \frac{5 - 4.8}{5} \times 100 = 4\%$$

So there is an error of about 4% in verification of superposition theorem

This may have happened due to the following reasons:-

(i) Internal resistance of connecting wires.

(ii) Minor fluctuations in reading of instruments while doing the experiment

(iii) Increase in resistance of wires due to heating effect.

(iv) The value of resistance of resistors is not accurate

PART-B :- THEVANIN'S THEOREM

$$(V_{\text{TH}})_{\text{Theoretical}} = 6\text{V}$$

$$(V_{\text{TH}})_{\text{Experimental}} = 5.9\text{V}$$

$$\% \text{ error} = \frac{6 - 5.9}{6} \times 100 = 1.67\%$$

$$(R_{\text{TH}})_{\text{Theoretical}} = 2100\Omega$$

$$(R_{\text{TH}})_{\text{Practical}} = 2107\Omega$$

$$\% \text{ error} = \frac{2107 - 2100}{2100} \times 100 = 0.33\%$$

The difference in values might have been caused due to:-

- (i) Internal resistance of connecting wires.
- (ii) Incorrect resistance value of resistors
- (iii) Instruments not working properly
- (iv) Heating effect of electricity.

PART C:- MAXIMUM POWER TRANSFER THEOREM

The maximum power is consumed when the value load resistance is closest to the thevanin resistance. The power consumed at $R = 2.2 \text{ k}\Omega$ is equal to 4 mW and thus confirms the maximum power transfer theorem.

Conclusion:-

Due to very small errors (negligible) the superposition theorem, thevanin theorem and maximum power transfer theorems are verified.

PART D:- ANSWERS TO QUESTIONS

(1) In part A we got a 1% error which is very small. The errors are caused due to heating effects, incorrect value of readings in instruments etc. Otherwise the theorem is very much accurate.

(2) Difference in values of $V_{Th} = 0.14$ and $R_{Th} = 7 \Omega$. The difference can be due to various facts like error in instruments, heating effects, incorrect value of resistance etc. (yes or no)

(3) $R_L = 560 \Omega$ - Power Consumed - 2.83 mW
 $1 \text{ k}\Omega$ - 3.61 mW
 $2.2 \text{ k}\Omega$ - 4 mW
 $3.9 \text{ k}\Omega$ - 3.9 mW
 $4.7 \text{ k}\Omega$ - 3.4 mW

We can observe power increases till $2.2 \text{ k}\Omega$ then it decreases. The theoretical maximum for power transfer is at $2.1 \text{ k}\Omega$ when $R_L = R_{Th}$. $2.2 \text{ k}\Omega$ is very close to $2.1 \text{ k}\Omega$ compared to other resistance values. Thus we can say maximum power transfer theorem holds.